

3D Vertical FeFET Array with Record Endurance ($>10^{12}$), Fast Writing ($\pm 2V$, 20 ns), Disturb Immunity, and Kb-scale Verification for High Density 1T RAM

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Abstract

We report a comprehensively optimized 3D vertical AND-type ferroelectric FET (FeFET) memory, with highlights below: 1) Record-level key metrics of the 3D cells, including endurance $>10^{12}$, fast write at $\pm 2V/20$ ns, immediate read-after-write, and $0.001\ \mu\text{m}^2$ channel area; 2) First demonstration of 3D array reliability, showing disturb immunity with aggressively scaled 40-nm vertical pitch, and error-free Kb-scale readout; 3) Systematic process modulation with in-depth nano-analysis, revealing co-optimization mechanisms related to FE grain and channel oxygen engineering. The proposed 3D FeFETs enable high density 1T RAM, offering a low-cost and high-speed solution for the expanding hot data in AI era.

Introduction

The rapid evolution of generative AI accelerates the demand for fast RAM with high density and energy efficiency. However, conventional 1T1C DRAM faces scaling bottlenecks and the refresh power overhead (Fig. 1(a)) [1]. Ferroelectric (FE) FET (FeFET) has thus emerged as a compelling memory, leveraging the 1T architecture for high-density 3D vertical integration and FE's inherent non-volatility (Fig. 1(b-c)) [2]. Synergizing with DRAM, the 3D FeFET enables a cost-and-energy efficient technology for the growing hot data, such as KV cache in LLMs. Yet, practical deployment of 3D FeFET remains hindered by its limited endurance, slow write speed, and various disturbs in 1T array [3]. This work effectively addresses the above issues and demonstrates a Kb-scale, 4-layer 3D-AND FeFET array. The proposed FE superlattice and channel oxygen vacancy engineering enable fast erase and prevent defects generation at FE interface. Array disturbs are minimized by co-optimizing coercive field (E_c) distribution and operation schemes. Physical origins of the improvement are revealed via nano-characterization, and potential for high-density 1T RAM is validated by TCAD simulation.

Fabrication and Characterization of 3D FeFETs

The 3D FeFETs are fabricated via the process flow in Fig. 2(d). Both the HZO FE and IGO channel (Ch.) are grown via atomic layer deposition (ALD), with various process conditions for comparison. SEM and HRTEM images in Fig. 2(a-e) confirm the successful fabrication of a 4-layer 32×32 array, featuring an aggressively scaled 40 nm vertical pitch and a 50 nm channel length (L_{ch}). The side-wall devices show 6 nm HZO (solid solution (SS) or superlattice (SL)) with 4 nm IGO channels (O-rich or O-poor), consistent with design (Fig. 2(f)).

The DC I_d - V_g curves of different FeFETs are displayed in Fig. 3(a). All devices show memory window (MW) of $\sim 0.7V$, with large I_{on}/I_{off} of $>10^5$. Devices across the four layers exhibit nearly identical I_d - V_g curves (Fig. 3(b)), confirming minimal layer-to-layer variation. Under pulse operations, stable $\sim 0.65V$ MW is maintained (Fig. 3(c)), with negligible read-after-write-delay (RAWD) (Fig. 3(d)). To investigate the write property, pulses with various amplitudes and widths are applied for MW mapping (Fig. 4). While the O-rich-Ch devices require $\sim 2.5V/50$ ns to erase, the SL+O-poor-Ch device achieves a faster $\pm 2V$, 20 ns switching. This accelerated erase speed is attributed to higher oxygen vacancy (V_o) concentration in O-poor channel, which facilitates minority carrier generation and allows more efficient field-driven FE switching.

The write endurance of the FeFETs is measured in Fig. 5(a) and (b). The endurance of SS+O-rich-Ch is 10^{11} cycles, which slightly increases to 2×10^{11} for SL+O-rich-Ch. The SL+O-poor-Ch configuration exhibits superior endurance of $>10^{12}$, where the MW remains $>0.3V$. Read endurance is tested

to $>10^{12}$ cycles with no V_{th} shift, demonstrating highly reliable readout (Fig. 5(c)). The FeFETs show stable retention for up to 3000 s, extrapolated to >10 years (Fig. 5(d)). To investigate the mechanisms underlying endurance optimization, STEM-EELS characterization is performed on the FE layers of all FeFETs (Fig. 6(a-b)). The B/A peak ratio in O-K edge spectroscopy signifies the V_o concentration. Fig. 6(c) quantifies the B/A difference between pristine and cycled devices, indicating the least V_o generation in SL+O-poor-Ch device. The stabilized interface by SL HZO and the reduced erase stress by O-poor channel suppressed interface V_o generation, leading to higher endurance (Fig. 6(d)).

Optimization of Write Disturbs in 3D FeFET Array

Fig. 7(a) depicts the conventional write scheme of 3D FeFET array, showing the $V_w/2$ (half-selected) and cell-to-cell (C2C) disturb. The origin and solutions of the two disturbs are illustrated in Fig. 7(b-c). To suppress $V_w/2$ disturb, engineering of O-pulse time in HZO ALD growth is employed, with results shown in Fig. 7(d-e). While the O-1.6s device exhibits notable V_{th} shift, the O-0.1s remains unaffected after 10^9 disturb cycles. Fig. 8(a) shows the effect of $V_w/2$ stress in FE capacitors, also demonstrating improved disturb immunity in O-0.1s. HRTEM images in Fig. 8(b) indicates a larger grain size in O-0.1s HZO. The average grain diameter is extracted from plan-view SEM, further evidencing the larger grain size in O-0.1s (Fig. 8(c-d)). Fig. 8(d) illustrates the relation between grain size and disturb property: larger grain sizes result in a tighter E_c distribution, reducing the FE domains flipped by residue V_w . For C2C disturb, a novel write scheme is proposed, which reduces the stress between adjacent cells from $V_w/2$ to $V_w/3$ (Fig. 9(a)). Although it increases stress on specific half-selected cells to $2V_w/3$, Fig. 9(b) proves they remain immune to such disturbs over 10^9 cycles. Overall, the proposed scheme reduces C2C disturb by up to 43%, limiting V_{th} shift to <50 mV at 40 nm pitch (Fig. 9(c-d)), which is smaller than the most-scaled vertical pitch (42 nm) reported in 3D NAND and FeNAND [4]. Experimental validation on a $4\times 16\times 16$ array confirms good uniformity and Kb-scale error-free operation (Fig. 10).

Simulation and Benchmark

TCAD and SPICE simulations are conducted to investigate the integration potential of the 3D vertical FeFETs, with the structure and key parameters presented in Fig. 11(a-c). The architecture exhibits competitive read latency. Based on cost-scalable vertical stacking, the bit density can reach >10 times of 1b DRAM (Fig. 11(d-e)). As benchmarked in Table 1 [2, 3, 5-7], Our 3D FeFET achieves record high endurance, fast write speed, low operation voltage, fully demonstrated array disturb immunity, and scalable bit density compared to state-of-the-art FE-based memories.

Conclusion

This work demonstrates a Kb-scale, 4-layer 3D vertical FeFET array with 40 nm vertical pitch. Via synergistic HZO/channel engineering and a novel write scheme, we achieve fast write at $\pm 2V/20$ ns, record-high 10^{12} endurance, disturb immunity, and minimal RAWD. Physical mechanisms are revealed by nano-analysis, guiding further optimizations. The 3D vertical FeFETs show strong potential for future high-speed RAM technology with scalable bit-density.

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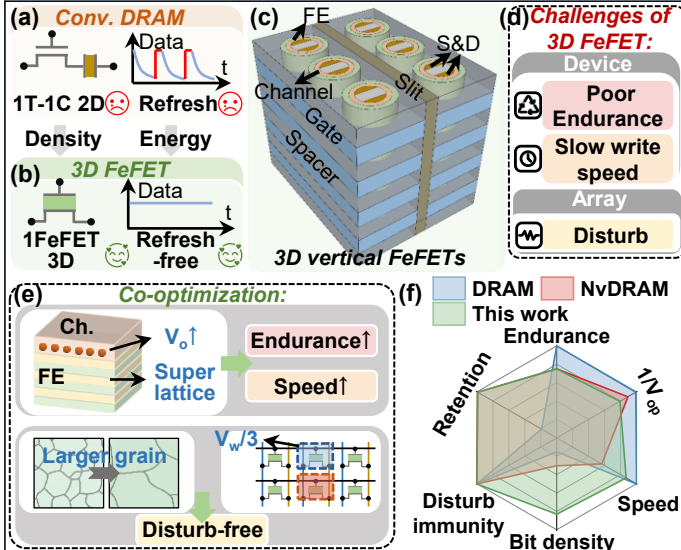


Fig. 1 (a) and (b) Comparison of conventional 1T1C DRAM and 3D FeFET with high density and low energy consumption. (c) Structure schematic of 3D vertical FeFETs. (d) Key challenges of 3D FeFET. (e) Co-optimization to address the challenges. (f) Comparison of DRAM, NvDRAM and this work.

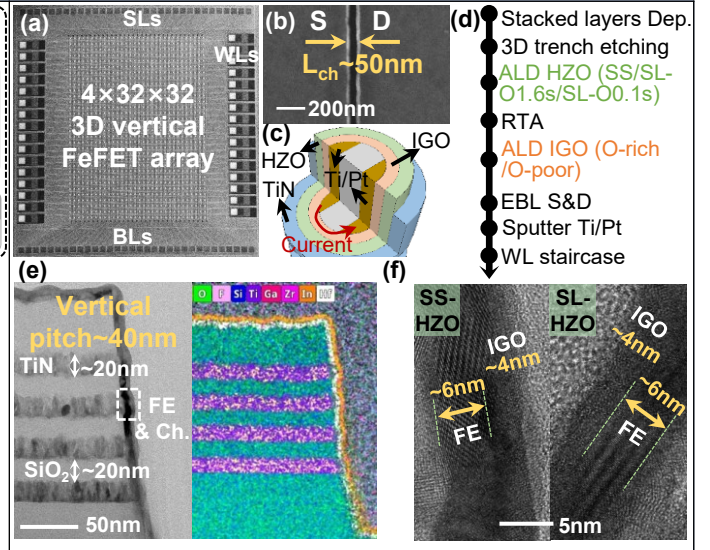


Fig. 2 (a) and (b) SEM images of the fabricated 3D FeFET array. (c) Schematic and (d) process flow of the 3D FeFETs. (e) Cross-sectional TEM and EDS mapping of the 3D FeFET, demonstrating highly scaled vertical pitch of 40nm. (f) HRTEM images of SS-HZO and SL-HZO with IGO channel.

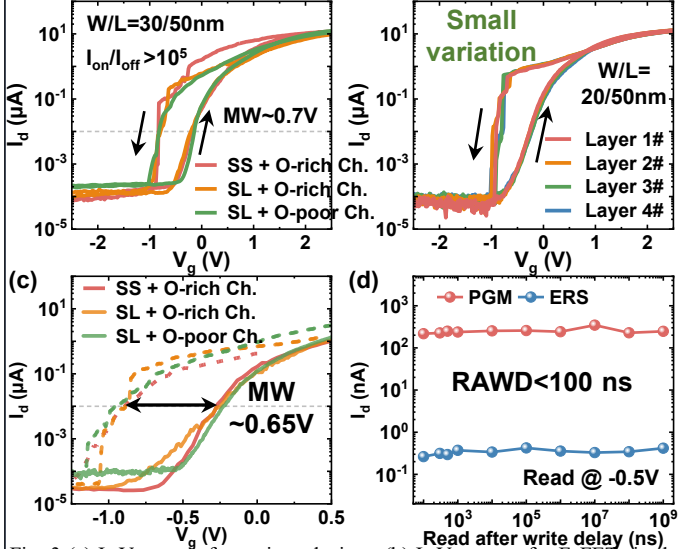


Fig. 3 (a) I_d - V_g sweep for various devices. (b) I_d - V_g curves for FeFETs in the four layers, showing small layer-to-layer variation. (c) I_d - V_g curves after pulse writing for various devices. All FeFETs show MW of $\sim 0.65V$. (d) Read-after-write-delay of the FeFETs, with minimal read latency of < 100 ns.

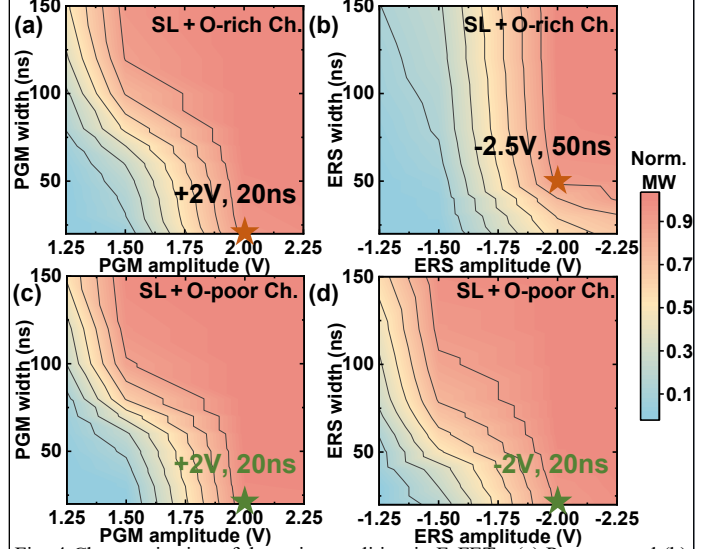


Fig. 4 Characterization of the write condition in FeFETs. (a) Program and (b) erase condition of SL+O-rich FeFET, which is +2 V, 20 ns and -2.5V, 50 ns, respectively. (c) Program and (d) erase condition of SL+O-poor device. Effective writing can be achieved with only $\pm 2V$ and 20 ns.

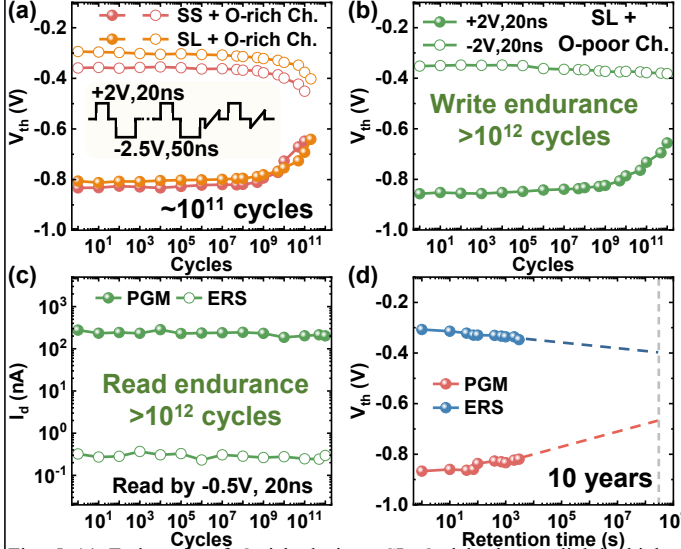


Fig. 5 (a) Endurance of O-rich devices. SL+O-rich shows slightly higher endurance of 2×10^{11} cycles. (b) SL+O-poor FeFET demonstrates the highest endurance of $> 10^{12}$ cycles. (c) Read endurance of $> 10^{12}$ cycles, confirming reliable readout. (d) Retention of FeFET, extrapolated to 10 years.

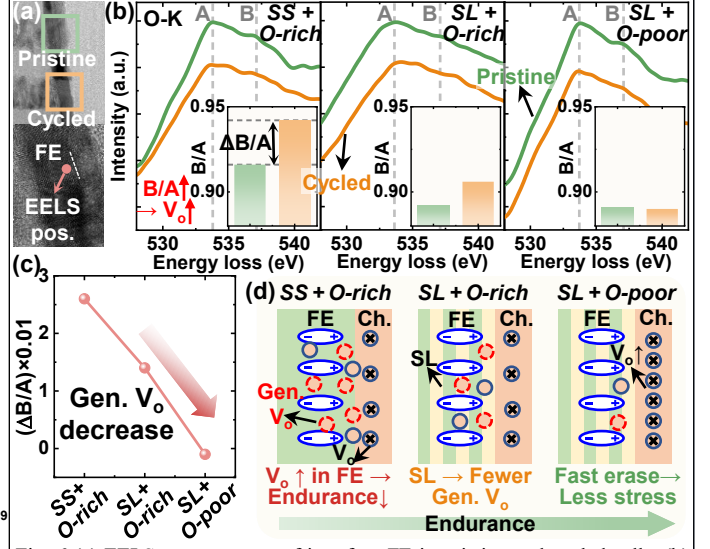


Fig. 6 (a) EELS measurement of interface FE in pristine and cycled cells. (b) EELS O-K edge curves, where V_o concentration is reflected from B/A peak ratio. (c) Comparison of the increased V_o during cycling, which is minimized in SL+O-poor device. (d) Mechanisms of the endurance improvement.

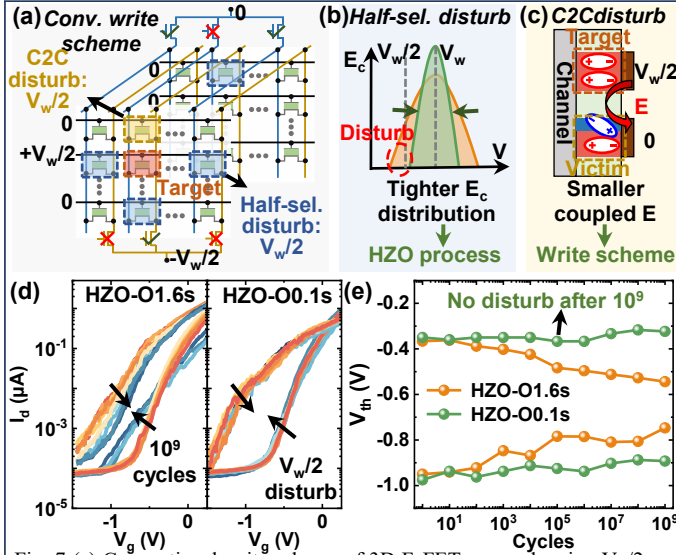


Fig. 7 (a) Conventional write scheme of 3D FeFET array, showing $V_w/2$ and cell-to-cell (C2C) disturb. (b) and (c) Origins and solutions of the disturb. (d) and (e) Cumulative effect of $V_w/2$ disturb in different HZO processes. HZO with O pulse of 0.1 s show nearly no disturb after 10^9 cycles.

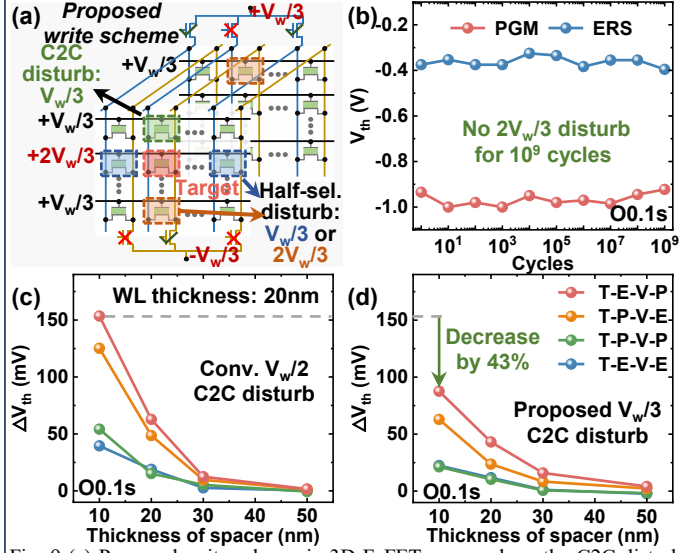


Fig. 9 (a) Proposed write scheme in 3D FeFET array, where the C2C disturb stress decreases from $V_w/2$ to $V_w/3$. (b) No Half-select disturb observed even with stress increased to $2V_w/3$. (c) and (d) Verification of the C2C disturb reduction. The V_{th} shift decreases by 43% in our proposed write scheme.

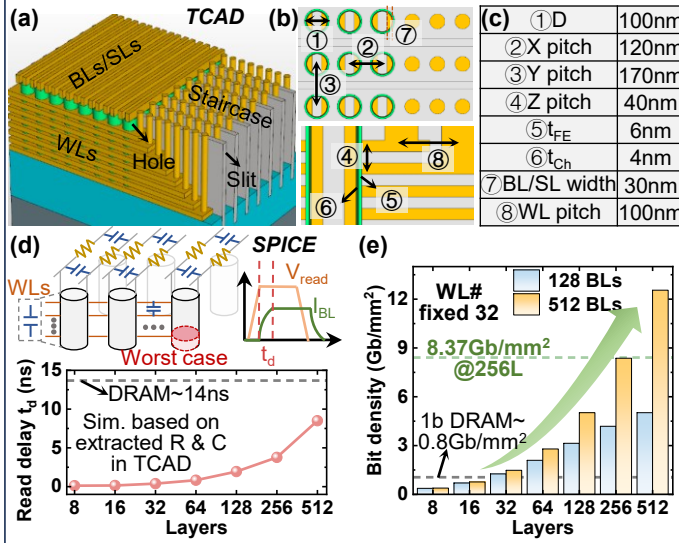


Fig. 11 (a) Simulation of the 3D vertical FeFET array in sentaurus TCAD. (b) Plan-view and cross-sectional layout. (c) Key parameters in simulation. (d) Simulated read latency across various layers of 3D FeFET in SPICE. (e) Simulated bit density of the 3D FeFET (WL contact area included).

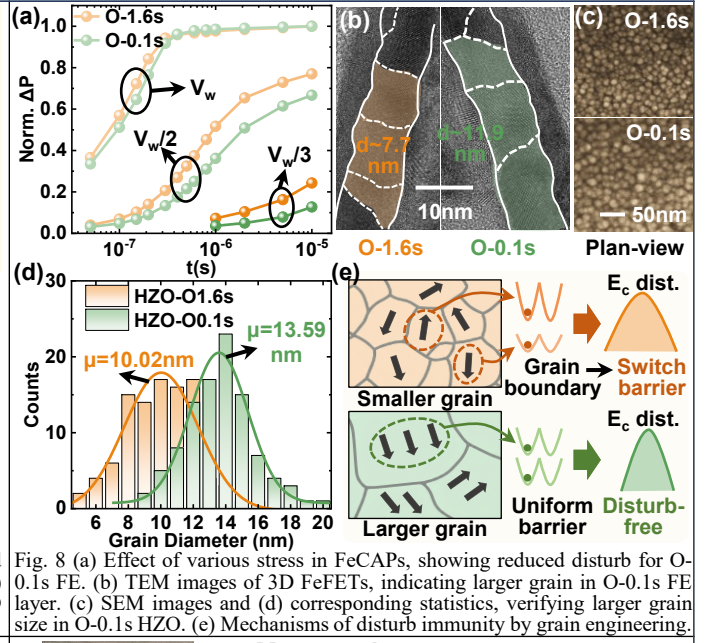


Fig. 8 (a) Effect of various stress in FeCAPs, showing reduced disturb for O-0.1s FE. (b) TEM images of 3D FeFETs, indicating larger grain in O-0.1s FE layer. (c) SEM images and (d) corresponding statistics, verifying larger grain size in O-0.1s HZO. (e) Mechanisms of disturb immunity by grain engineering.

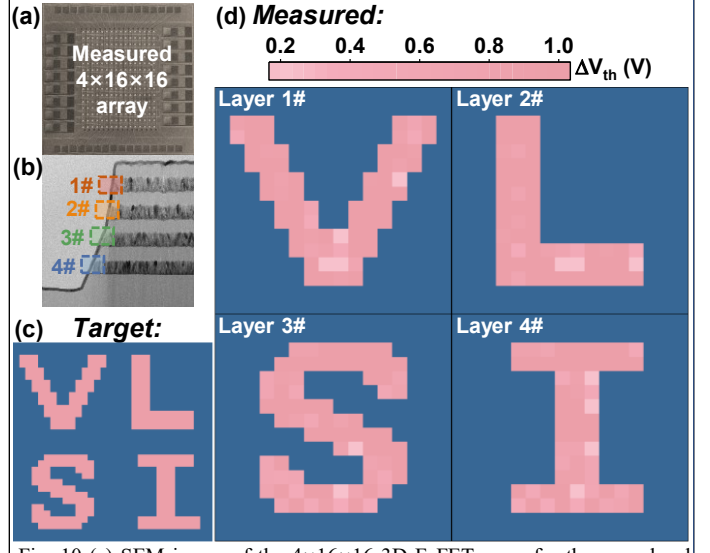


Fig. 10 (a) SEM image of the 4x16x16 3D FeFET array for the array-level test. (b) TEM image of the FeFET array, showing 4 stacked layers to measure. (c) Target pattern for writing in the array. (d) Measured ΔV_{th} of the FeFETs in four layers, demonstrating good uniformity and error-free results.

Table 1. Benchmark of the FE-based RAM

	IEDM23 [5]	IEDM25 [6]	IEDM25 [7]	VLSI24 [3]	VLSI25 [2]	This work
Cell	1T1C	1T1C	1FeFET	1FeFET	1FeFET	1FeFET
Arch.	3D sequential	3D parallel ^a	2D	3D vertical	3D vertical	3D vertical
Voltage	1.5V	2V	3V	4V	3.25V	2V
Speed	100ns	-	20ns	200ns	50ns	20ns
Endurance	>10 ¹²	10 ⁸	10 ¹²	10 ⁷	10 ¹¹	>10 ¹²
Retention	10 yrs	10 yrs	10 yrs	10 yrs	10 yrs	10 yrs
Disturb immune	-	-	-	-	-	Half-sel. & C2C
Array demo.	32Gb (2 layers)	-	No	(3 layers)	8x8 (3 layers)	4kb (4 layers)
Bit density	0.45 Gb/mm ²	~3.0 ^b Gb/mm ²	-	-	-	8.37 ^c Gb/mm ²

^a3D DRAM-like. ^bSimulated in 100 layers. ^cSimulated for 256 layers.

Reference [1] K. S. Choi *et al.*, *Symp. on VLSI*, 2024, pp. 1-2, doi:10.1109/VLSITechnologyandCir46783.2024.10631471. [2] Y. Zhou *et al.*, *Symp. on VLSI*, 2025, pp. 1-3, doi: 10.23919/VLSITechnologyandCir65189.2025.11074820. [3] Y. Feng *et al.*, *Symp. on VLSI*, 2024, pp. 1-2, doi: 10.1109/VLSITechnologyandCir46783.2024.10631352. [4] P. Sharma *et al.*, *Symp. on VLSI*, 2025, pp.1-3, doi: 10.23919/VLSITechnologyandCir65189.2025.11074815. [5] N. Ramaswamy *et al* *IEDM*, 2023, pp. 1-4, doi: 10.1109/IEDM45741.2023.10413848. [6] H. Li *et al.*, *IEDM*, 2025, pp. 1-4. [7] J.-K. Kim *et al.*, *IEDM*, 2025, pp. 1-4.